

Evan J. Turner

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EDUCATION

University of Wisconsin-Madison, Madison, WI

Master of Science, Mechanical Engineering

Expected Dec 2026

Clarkson University, Potsdam, NY

Bachelor of Science with Honors, Mechanical Engineering (Minor: Biomedical Engineering)

May 2024

Leadership: President & Captain, Men's Club Ice Hockey Program

RELEVANT EXPERIENCE

Graduate Research Assistant, Cardiovascular Fluid Dynamics Lab (UW-Madison)

Aug 2024 – Present

- Evaluated wall shear stress and velocity metrics' ability to identify heightened risk pregnant patient groups using doppler US derived low-fidelity computational fluid dynamics (CFD).
- Modeled patient-specific Fontan hemodynamics via physiologic CFD to predict energy loss and flow distribution for surgical planning.
- Characterized nephrostomy tube blockage from exchange procedures by designing an experimental setup that established quantitative diagnostic metrics.
- Assessed urethral biomechanics in stricture patients by developing a non-invasive MRI and CFD framework.

Biomedical Engineering Co-op, ZOLL Medical Corporation

Jan 2023 – Aug 2023

- Developed novel polymer fabrication process, increasing process reliability from <5% to almost 100%, boosting fabrication speed by ~40x and device sensitivity by 10x.
- Prototyped mechanical test fixtures by modeling systems in SolidWorks and manufacturing components via 3D printing and laser cutting.
- Designed and executed engineering studies to verify new product performance using custom-built MATLAB signal processing tools for time-frequency data analysis.

Research Assistant, Connective Tissue Mechanics Lab (Clarkson University)

May 2021 – May 2024

- Developed an ex-vivo model to quantify post-retraction intradiscal injection leakage, establishing a benchmark for rapid risk factor screening.
 - Led a study using the technique to investigate post-procedure movement as a leakage risk factor.
- Designed a low-cost, wearable spinal motion tracking system utilized across multiple biomechanics studies, including investigation of spinal motion coordination post-fatigue in competitive Nordic skiers.

Makerspace Supervisor, Dorf Makerspace

Sep 2020 – May 2024

- Maintained and calibrated equipment including FDM & SLA 3D printers, laser cutters, CNC routers, and soldering stations to ensure safe and effective operation.
- Instructed students and faculty through hands-on software and tool workshops, daily equipment demos, and general design discussions to help users bring their ideas to life.

Mechanical Engineering Intern, ABTech, Inc.

May 2022 – Aug 2022

- Engineered ultra-precision metrology and motion systems using Design for Manufacturing (DFM) principles, executing CAD modeling, hands-on machine shop fabrication, and electro-mechanical assembly.
- Designed a custom probe mounting fixture for an engine assembly system and reverse-engineered competitor air-bearing spindles to capture new servicing opportunities.

TECHNICAL SKILLS

- **Computer Aided Design:** SolidWorks (CSWA for Mechanical Design), AutoCAD, Fusion360, KiCAD
- **Programming & Simulation:** MATLAB, Python, C++, LabVIEW, ANSYS, Altair, SimVascular
- **Manufacturing:** 3D Printing, Laser Cutting, CNC Routing, Soldering, Basic Lathe, General Shop Equipment

PORTFOLIO & SELECTED WORKS

- **Project Portfolio:** [evanjturner.github.io](https://github.com/evanjturner) (Detailed explanations of school & personal projects)
- **Publications:** "Establishing a model for assessing IVD post-injection leakage", European Spine Journal (2022)
- **Presentations:** "WSS-Based Differentiation of Hypertensive Disorders Using US-Based CFD", ASME SBC (2025) | "MRI-Based CFD in Urethral Stricture Biomechanics", ISMRM Annual Meeting (2025)